STRING MATCHING APPLICATION USING THE Z-ALGORITHM

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ABSTRACT

Our project aims to develop a document similarity checking application using the Z-Algorithm. The application compares the textual content of two PDF documents to identify matching or duplicated content. By extracting text from the uploaded PDFs and applying efficient string-matching techniques, the system can determine the similarity between the documents and display the matching portions.

PROBLEM STATEMENT

Comparing large project documents manually to identify copied or duplicated content is time-consuming and difficult.

Students may copy another student's project report while changing only details such as the name or roll number.

Therefore, an efficient system is required to compare two documents, identify matching textual content, and determine their similarity.

PROPOSED FUNCTIONALITIES

1. 📄 PDF Upload
Upload two PDF documents for comparison.

2. 📝 Text Extraction
Extract textual content from both PDFs.

3. 🔍 Z-Algorithm Based Matching
Compare the extracted text using the Z-Algorithm.

4. 📊 Similarity Calculation
Calculate the percentage of matching content.

5. 🖥️ Visual Results
Display the similarity percentage and matching portions.

TOOLS AND TECHNOLOGIES

TOOLS	TECHNOLOGIES
Z-Algorithm	Core string-matching algorithm
PDF Text Extraction	Extract text from uploaded PDFs
Document Comparison	Compare extracted text and identify matching content
User Interface	Upload PDFs and display similarity results

EXPECTED OUTCOME

- **Compare** two PDF documents and identify similar content.
- **Generate** a similarity percentage based on the matching text.
- **Identify** and display matching portions of the documents.
- **Efficiently** detect copied or duplicated content using Z-Algorithm-based string matching.

Expected Result: A simple and efficient document similarity-checking application.


GPS Map Camera
Hyderabad,Telangana,India
Dollar Dreams, Bowrampet, Hyderabad, Telangana 500043, India
Lat 17.547206, Long 78.404749
Thursday, 20/08/2026 01:30 PM GMT+05:30
Note : Captured by GPS Map Camera

THANK YOU!